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Degree: B.Tech

Roll no.:
Semester: 3

Mid Semester Examination, (Summer) June 2026

Course Code: ITMTC301
Course title: Optimization Principles and Techniques

Time: 1 hour 30 minutes

Maximum Marks. 20

Note: Attempt all the questions in given order only. Missing data/information(if any), may be suitably assumed and mentioned in the answer.

Q. No.	Questions	Marks	CO																														
1a	Solve the given LPP graphically: Max Z = $-2x - 4y$ subject to $-x_1 + x_2 \leq 1,$ $2x_1 - x_2 \leq 2,$ $x_1, x_2 \geq 0.$	2	CO1																														
1b	Identify all basic feasible solutions for the system. $x_1 + 4x_2 + x_3 = 8,$ $x_1 + 2x_2 + x_4 = 4,$ $x_1, x_2, x_3, x_4 \geq 0.$	2	CO1																														
2a	Solve the given LPP using simplex method: Max Z = $10x_1 + 6x_2$ subject to $5x_1 + 3x_2 \leq 30$ $x_1 + 2x_2 \leq 18$ $x_1, x_2 \geq 0$	2	CO1																														
2b	A manufacturing company makes two models A and B of a product. Each piece of Model A requires 9 labour hours for fabricating and 1 labour hour for finishing. Each piece of Model B requires 12 labour hours for fabricating and 3 labour hours for finishing. For fabricating and finishing, the maximum labour hours available are 180 and 30 respectively. The company makes a profit ofRs 8000 on each piece of model A and Rs 12000 on each piece of Model B. How many pieces of Model A and Model B should be manufactured per week to realise a maximum profit? Formulate this problem as LPP.	2	CO1																														
3a	The following table shows the information on the availability of supply to each warehouse, the requirement of each market and unit transportation cost (in Rs.) from each warehouse to each market. <table><tr><td></td><td>M1</td><td>M2</td><td>M3</td><td>M4</td><td>Supply</td></tr><tr><td>W1</td><td>6</td><td>3</td><td>5</td><td>4</td><td>22</td></tr><tr><td>W2</td><td>5</td><td>9</td><td>2</td><td>7</td><td>15</td></tr><tr><td>W3</td><td>5</td><td>7</td><td>8</td><td>6</td><td>8</td></tr><tr><td>Demand</td><td>7</td><td>12</td><td>17</td><td>9</td><td></td></tr></table> Solve the given transportation problem using Vogel's Approximation Method.		M1	M2	M3	M4	Supply	W1	6	3	5	4	22	W2	5	9	2	7	15	W3	5	7	8	6	8	Demand	7	12	17	9		2	CO2
	M1	M2	M3	M4	Supply																												
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W3	5	7	8	6	8																												
Demand	7	12	17	9																													

3b	Write the dual of the following LPP: Max Z = $7x_1 + 8x_2$ subject to $4x_1 + 8x_2 - 3x_3 \geq 9$ $5x_1 + 2x_2 + 8x_3 \leq 5$ $x_1 + 2x_2 + 2x_3 = 10$ $4x_1 + 3x_2 + x_3 \geq 7$ $x_1 \geq 0, x_2$ unrestricted, $x_3 \leq 0$	2	CO2																																				
4a	We have to assign five jobs, I, II, III, IV, V to five workers A, B, C, D, and E. The time taken by different workers (in hours) in completing different jobs is given below: <table><tr><td></td><td>I</td><td>II</td><td>III</td><td>IV</td><td>V</td></tr><tr><td>A</td><td>85</td><td>75</td><td>65</td><td>125</td><td>75</td></tr><tr><td>B</td><td>90</td><td>78</td><td>66</td><td>132</td><td>78</td></tr><tr><td>C</td><td>75</td><td>66</td><td>57</td><td>114</td><td>69</td></tr><tr><td>D</td><td>80</td><td>72</td><td>60</td><td>120</td><td>72</td></tr><tr><td>E</td><td>76</td><td>64</td><td>56</td><td>112</td><td>68</td></tr></table> Find the minimum time (in hours) taken by the workers to complete all jobs.		I	II	III	IV	V	A	85	75	65	125	75	B	90	78	66	132	78	C	75	66	57	114	69	D	80	72	60	120	72	E	76	64	56	112	68	2	CO2
	I	II	III	IV	V																																		
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C	75	66	57	114	69																																		
D	80	72	60	120	72																																		
E	76	64	56	112	68																																		
4b	Solve the following LPP using dual simplex method: Min Z = $5x_1 + 6x_2$ subject to $x_1 + x_2 \geq 2$ $4x_1 + x_2 \geq 4$ $x_1 \geq 0, x_2 \geq 0$	2	CO2																																				
5a	Find the initial basic feasible solution of the given transportation problem using Least cost method. <table><tr><td></td><td>M1</td><td>M2</td><td>M3</td><td>M4</td><td>Supply</td></tr><tr><td>W1</td><td>11</td><td>13</td><td>17</td><td>14</td><td>250</td></tr><tr><td>W2</td><td>16</td><td>18</td><td>14</td><td>10</td><td>300</td></tr><tr><td>W3</td><td>21</td><td>24</td><td>13</td><td>10</td><td>400</td></tr><tr><td>Demand</td><td>200</td><td>225</td><td>275</td><td>250</td><td></td></tr></table>		M1	M2	M3	M4	Supply	W1	11	13	17	14	250	W2	16	18	14	10	300	W3	21	24	13	10	400	Demand	200	225	275	250		2	CO2						
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Demand	200	225	275	250																																			
5b	Consider the LPP $\text{Max } 12x_1 + 12x_2$ subject to $x_1 + x_2 + x_3 = 4$ $x_1 + 4x_2 + x_4 = 10$ $x_1, x_2, x_3, x_4 \geq 0.$ The optimum table is given by: <table><tr><td>Basis</td><td>Solution</td><td>x_1</td><td>x_2</td><td>x_3</td><td>x_4</td></tr><tr><td>x_1</td><td>4</td><td>1</td><td>1</td><td>1</td><td>0</td></tr><tr><td>x_4</td><td>6</td><td>0</td><td>3</td><td>-1</td><td>1</td></tr><tr><td>z</td><td>48</td><td>0</td><td>0</td><td>12</td><td>0</td></tr></table> Write the dual problem and find the solution of the dual from the optimum primal table.	Basis	Solution	x_1	x_2	x_3	x_4	x_1	4	1	1	1	0	x_4	6	0	3	-1	1	z	48	0	0	12	0	2	CO2												
Basis	Solution	x_1	x_2	x_3	x_4																																		
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